two hydrogen atoms and one oxygen atom, instead of—let us say—four hydrogen and two oxygen atoms, which would still represent the same mass ratio? In order to answer this question, we must make reference again to the theory of gases, and to Avogadro's hypothesis in particular, which states—as mentioned above—that at the same temperature and pressure, equal volumes of all gases contain the same number of molecules. There exists an exact proof of this hypothesis, but here we shall confine ourselves merely to elucidating it. The pressure on the sides of a vessel filled with a gas, is due to the impacts of the gas molecules, which hammer against those sides like drops of a steady rain, only to be bounced back from them again. The sum total of the forces of these impacts exerts a pressure against the sides. The magnitude of this pressure, obviously, depends on the kinetic energy of the molecules, which, in turn, depends on the temperature of the gas. Since the days of Maxwell, we have known that at the same temperature the molecules of all gases always have the same average kinetic energy. Thus, if equal volumes of gases contain the same number of molecules, it follows that at the same temperature their pressure, too, must be the same. This is the real meaning of Avogadro's hypothesis.

Moreover, at this point, we must refer to certain chemical facts. The fact, already mentioned, that 2 grammes of gaseous hydrogen and 16 grammes of gaseous oxygen combine to form 18 grammes of water vapour, can be expressed by the following equation:

$$\text{2 gm. of hydrogen} + \text{16 gm. of oxygen} = \text{18 gm. of water vapour.}$$

Instead of the ratio by mass, we may investigate also the ratio by volume present when hydrogen and oxygen combine at the same temperature. Experiments show, for instance, that 1 litre of hydrogen and $\frac{1}{2}$ litre of oxygen combine to form 1 litre of water vapour. Thus, we get the following equation:

$$\text{1 litre of hydrogen} + \tfrac{1}{2}\text{ litre of oxygen} = \text{1 litre of water vapour.}$$

From these two equations, it is easy to deduce the mass ratios of these three kinds of molecules. Since 1 litre always contains the same number of gas molecules, all we have to do is determine the ratio of the above-mentioned masses and volumes of the three gases, to obtain numbers proportionate to the masses of their molecules. Thus we get: